ECE 340 # Homework Assignment #2 **Fall 2026**

DUE: FRIDAY, SEPTEMBER 11, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN BEFORE 5:00PM ON THE DUE DATE. PER THE SYLLABUS LATE HOMEWORK IS NOT ACCEPTED.

1. CHARGE AND CARRIER ENERGY

Consider the band diagram below in Fig. 1.1:

(A) Which hole has the highest kinetic energy? Why?

(B) Which hole has the highest potential energy? Why?

(C) Which hole has the highest total energy? Why?

(D) Which of these valence band states has the highest probability to be occupied by an electron?

(E) Assuming the kinetic energy of the hole in region (3) is 2.5 eV, that the hole effective mass is $1.5\,m_0$, and that the electron effective mass in the conduction band is $0.75\,m_0$, what is the kinetic energy of an electron in the conduction band with the same momentum as this hole?

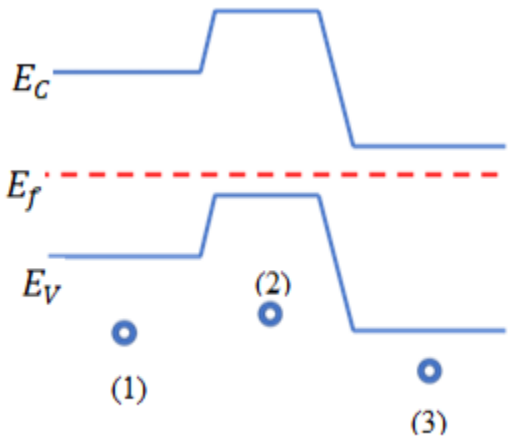


Figure 1.1: Energy Diagram for Q1

2. Materials and Band Structure

(A). Answer i and ii below:

i. Draw the $E-r$ (energy versus position) diagram for an insulator, a semiconductor, and a metal at 0 K. Label your diagram and show where the electrons reside.

ii. State the difference between the $E-r$ diagram of an insulator and a semiconductor at 0 K. How does the conduction band of a metal differ from those of an insulator and a semiconductor at 0 K?

(B). Consider the $E-k$ diagram of Fig. 2.1. The energy axis has units of eV. Answer (i)–(iv) based on this diagram.

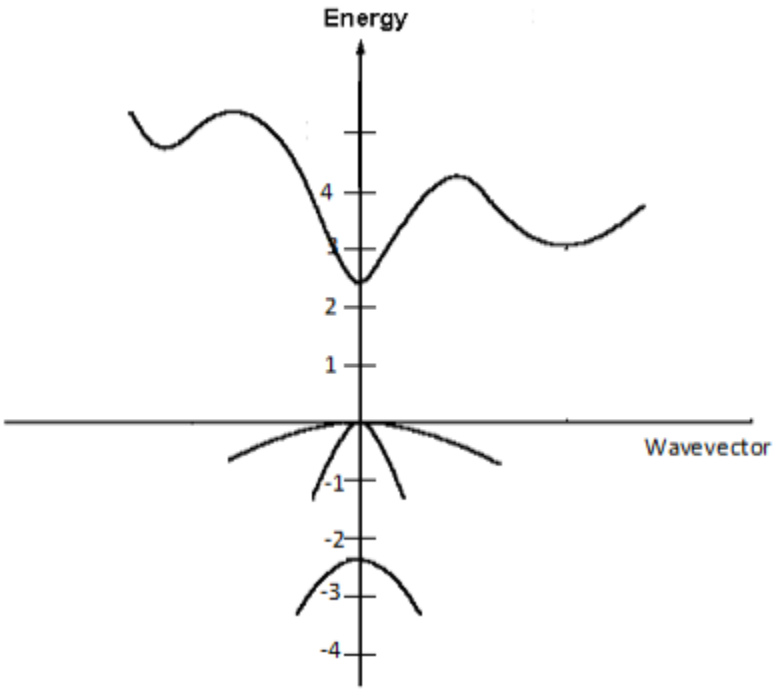


Figure 2.1: E–k diagram for Q2 (B–E)

i. Label each of the following bands on the diagram: conduction band (CB), light hole valence band (LHVB), heavy hole valence band (HHVB). State the reasons behind your labeling.

ii. Is this a direct or indirect bandgap material? Why? What is the value of the bandgap?

iii. An electron has a kinetic energy of 1.2 eV. Mark its position on the diagram.

iv. The curve near the conduction-band minima in the $E-k$ diagram follows the equation

$$E = 5 \times 10^{-38} k^2 \; Joules. \tag{2.1}$$

Find the electron effective mass.

3. Extrinsic and Intrinsic Materials

Silicon (Si) has an intrinsic carrier concentration $n_i = 1.5 \times 10^{10} cm^{-3}$ at 300 K.

(A). Assume the silicon sample is intrinsic at 300 K.

i. Find the electron concentration n.

ii. Find the hole concentration p.

iii. Briefly explain why $n = p$ in intrinsic material.

(B). Now the silicon is doped with acceptor atoms (B) at a concentration $N_A = 1.0 \times 10^{16} cm^{-3}$. Assume all acceptors are ionized, at thermal equilibrium, and no donor impurities.

i. Determine the majority carrier concentration.

ii. Calculate the minority carrier concentration using $np = ni^2$.

iii. Identify the majority and minority carriers.

(C). Explain with an energy band diagram and descriptive text why acceptor impurities can create free holes in the valence band at relatively low temperatures, while intrinsic carrier generation requires much higher temperatures.

4. Fermi–Dirac Statistics and Occupation Probability

(A). Calculate the probability of finding a filled electron state at the **conduction band edge** of an InP crystal ($E_g = 1.35\,eV$) at $T = 300\,\text{K}$ if the Fermi level (E_F) is at the following positions:

i. $E_F = E_C$

ii. $E_F = E_C - 6kT$

iii. $E_F = E_V$

(B). How does the probability of finding a filled electron state change (increase or decrease) as the Fermi level shifts from the conduction band towards the valence band in part (A)? Explain the reason behind this trend.

(C). Is the probability of finding an electron at the energy level $E_F + 1\,eV$ at 300 K higher or lower than at 100 K? Why?